ARM PrimeCell
MULTIPORT MEMORY CONTROLLER (PL172)
Errata Notice

ARM PrimeCell MULTIPORT MEMORY CONTROLLER PL172 Errata Notice

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General suggestion for additions and improvements are also welcome.

Contents

1 ABOUT THIS DOCUMENT

1.1 Change control

1.1.1 Change history

This refers to the revision changes to the Errata document

Issue	Date	Change
0.1	20 Dec 2001	Create template document
1.0	21 Dec 2001	First Release
2.0	12th June 2002	Updated for PL172r2p1-00rel0
2.1	22 July 2002	Updated for PL172r2p1-01rel0
2.2	26th Sept 2002	Updated to include 4 new erratas 3 on SDRAM accesses and 1 on arbitration
2.3	12th Nov. 2002	Added Errata 3.7
2.4	14 Nov. 2002	Added Errata 3.8
2.5	27 Nov 2002	Added Errata 3.9
3.0	3 Feb 2003	Release of PL172 r2p2-00ltd0
3.1	30 June 2003	Updated for patch release r2p2-00ltd1
3.2	12 May 2004	Updated defect 3.7

1.2 Errata List

This table summarises the errata notices detailed

Release: Release revision number affected

Section: Section in the document where the issues are described

Status: The status conditions are:

Open: Issue is current

Closed: Issue resolved in subsequent release

Severity: The severity conditions are:

A **Low** level of severity indicates that there is no direct impact to the silicon implementation. Examples are:

- No change to RTL required
- Documentation change
- Test pattern change
- Synthesis Scripts change

A **Medium** level of severity indicates that there is a minor impact to the silicon implementation, which can be overcome by modifications detailed in the errata notes. Examples are:

- A specific sequence of events not adhered to.
- Ambiguous or unexpected operation of the device.

A **High** level of severity indicates that there is a major impact to the silicon implementation, which cannot be overcome without modifications detailed in the errata notes. Examples are:

- Non conformance to the device specification.

Affected: Deliverables impacted by the issue

Description: Issue concerning Errata

Release	Section	Status	Severity	Affected	Description
1v0	2.1	Closed	Medium	RTL	Static Page-Mode Burst Reads (where HSIZE < Memory Data Bus Width)
1v0	2.2	Closed	Medium	RTL	Buffered Static Writes
1v0	2.3	Closed	Medium	RTL	Dynamic Memory CKE Deassertion during Auto-Refresh and Precharge-All
1v0	2.4	Closed	Medium	RTL	MPMC Locks-Up during Mixed Static Writes and Dynamic Accesses
1v0	2.5	Closed	Medium	RTL	MPMC Locks-Up during Mixed Static Reads and Dynamic Accesses
1v0	2.6	Closed	Low	Software	NAND Flash Control Software Drivers Not Included

1v0	2.7	Closed	Medium	Logs	Simulation Log Files Not Included
1v0	2.8	Closed	Low	UserGuide	Verification Simulations and Use of Denali Memory Models
1v0	2.9	Closed	Medium	RTL	Write-Read-Write Sequence in MPMCDynamic Memory
r2p1-00rel0	3.1	Closed	Low	Software	NAND Flash Control Software Drivers Not Included
r2p1-00rel0	3.2	Closed	Low	Verification	Makefiles missing
r2p1-00rel0	3.3	Closed	High	RTL	Data Lost During SDRAM Write Access
r2p1-00rel0	3.4	Closed	High	RTL	Wrong Data Access in Case of SDRAM Read
r2p1-00rel0	3.5	Closed	High	RTL	No SDRAM Refresh During Long Static Memory Access
r2p1-00rel0	3.6	Closed	High	RTL	Arbitration locking to a low priority AHB port in some circumstances
	3.7	Rejected			A Write Following a Burst Transfer is Ignored When No Wait States Are Used
r2p1-00rel0	3.8	Closed	Low	Software	Header file included that is not required
r2p1-00rel0	3.9	Closed	High	RTL	Data Read from memory and not from AHB write buffer if the buffers are full and the AHB write buffer does not contain a word of data
r2p2-00tld0	4.1	Open	High	RTL	CKE state at power-up.
r2p2-00ltd0	4.2	Closed	Low	Testbenches	An Obsolete Test Has Been Shipped and There are Issues with Some Tests and Log Files
r2p2-00tld0	5.1	Open	High	RTL	CKE state at power-up.

2 PL172 RELEASE 1V0 - ERRATA DETAILS

2.1 Static Page-Mode Burst Reads (where HSIZE < Memory Data Bus Width)

2.1.1 Summary

Static page-mode burst read accesses (of AHB width, HSIZE) from memory, where the external memory data bus is wider than HSIZE, return incorrect data.

2.1.2 Severity

Medium

This is a problem with the RTL but can be worked-around.

2.1.3 Symptoms

When performing a page-mode burst read from static memory, where the external memory data width is greater than HSIZE, the data returned for the first transfer in the burst is correct but all subsequent transfers within the burst return incorrect data.

2.1.4 Affected items

This is an issue with the RTL.

2.1.5 Short term work around

Any AHB reads from page-mode memory must be of HSIZE equal to or greater than the external memory data bus width.

If these accesses are unavoidable, the memory must be accessed as non-page-mode memory.

2.1.6 Proposed Corrective action

The RTL has been fixed in Release r2p1

2.2 Buffered Static Writes

2.2.1 Summary

Writes to static memory with the buffers enabled can fail in some cases.

2.2.2 Severity

Medium

This is a problem with the RTL but can be worked-around.

2.2.3 Symptoms

In some cases, writing to static memory with the buffers enabled can cause data to be written to incorrect addresses.

2.2.4 Affected items

This is an issue with the RTL.

2.2.5 Short term work around

When writing to static memory, the buffers must be disabled.

2.2.6 Proposed Corrective action

The RTL has been fixed in Release r2p1

2.3 Dynamic Memory CKE Deassertion during Auto-Refresh and Precharge-All

2.3.1 Summary

In some cases, CKE may deassert for a particular dynamic memory chip-select whilst the auto-refresh and precharge-all are being performed.

2.3.2 Severity

Medium

This is a problem with the RTL but can be worked-around.

2.3.3 Symptoms

During dynamic memory accesses incorrect data may be written or read. Since the memory device has not seen the precharge command, the previous row open would continue to be open. If the next access is to the same row, then there might not be a problem. But if the access is to a different row, then that particular row would not be open and that would result in a wrong read or write occurring.

2.3.4 Affected items

This is an issue with the RTL.

2.3.5 Short term work around

If this is an issue, the clock enables (CKE) may be set to be always on whilst accessing dynamic memory by setting the CE bit to 1 in the MPMCDynamicControl register.

2.3.6 Proposed Corrective action

The RTL has been fixed in Release r2p1

2.4 MPMC Locks-Up during Mixed Static Writes and Dynamic Accesses

2.4.1 Summary

A combination of static writes and dynamic accesses can cause the memory controller to lock-up.

2.4.2 Severity

Medium

This is a problem with the RTL but can be worked-around.

2.4.3 Symptoms

All HREADY signals go low, preventing further memory accesses.

The problem can be caused by a dynamic memory access followed by a static memory buffer-disabled write, with external memory data bus width greater than HSIZE. This static write is then followed by a dynamic memory access. If this happens, the memory controller locks-up.

2.4.4 Affected items

This is an issue with the RTL.

2.4.5 Short term work around

Any AHB writes to static memory must be of HSIZE equal to or greater than the external memory data bus width.

If this is unavoidable, the static writes should not occur at the same time as dynamic accesses.

2.4.6 Proposed Corrective action

The RTL has been fixed in Release r2p1

2.5 MPMC Locks-Up during Mixed Static Reads and Dynamic Accesses

2.5.1 Summary

A combination of static reads and dynamic accesses can cause the memory controller to lock-up.

2.5.2 Severity

Medium

This is a problem with the RTL but can be worked-around.

2.5.3 Symptoms

All HREADY signals go low, preventing further memory accesses.

The problem can be caused when there is a buffer-disabled static memory read of INCR4/8/16 or WRAP4/8/16 HBURST starting from a quad-word unaligned address and the dynamic access starts during the last access of the AHB burst. If this happens, the memory controller locks-up.

2.5.4 Affected items

This is an issue with the RTL.

2.5.5 Short term work around

Any AHB reads from static memory must be quad-word aligned.

If this is unavoidable, the static reads should not occur at the same time as dynamic accesses.

2.5.6 Proposed Corrective action

The RTL has been fixed in Release r2p1.

2.6 NAND Flash Control Software Drivers Not Included

2.6.1 Summary

The software drivers for NAND Flash are not included in Release 1v0.

2.6.2 Severity

Low

This is missing software that is not required for normal operation.

2.6.3 Symptoms

It will not be possible to use NAND Flash as standard at the current time.

2.6.4 Affected items

This is an issue with the software drivers.

2.6.5 Short term work around

It will not be possible to use NAND Flash as standard at the current time.

2.6.6 Proposed Corrective action

This product will not be able to support NAND Flash and thus there is no proposed corrective action.

2.7 Simulation Log Files Not Included

2.7.1 Summary

No log files from verification or integration tests are included in Release 1v0.

2.7.2 Severity

Medium

The simulations have been run by ARM to verify the design however no logs are available.

2.7.3 Symptoms

This will prevent the customer checking their simulation results against release logs.

2.7.4 Affected items

This affects checking of all simulation results.

2.7.5 Short term work around

It will not be possible to check results against reference log files.

2.7.6 Proposed Corrective action

Log files are included in release r2p1

2.8 Verification Simulations and Use of Denali Memory Models

2.8.1 Summary

There are multiple verification test benches are provided with this release of PL172. The standard make commands as detailed in the user guide will not start the verification simulations.

Different makefiles are required to run verification simulations.

2.8.2 Severity

Low

A different command must be used to start simulations

2.8.3 Symptoms

The verification simulations will fail to start.

2.8.4 Affected items

All verification simulations.

2.8.5 Short term work around

There are now multiple test benches available for use with the integration tests. These allow multiple configurations of memory models to be included in the verification tests. The integration test must be matched to the test being executed according to the table in section 8 of the Design Manual.

Make files are included in the verification directory to enable the test bench to be selected. These are named denali.mk and denali_LDV.mk.

For modeltech rtl simulations use the command:

Make –f denali.mk TBENCH=tbenchx rtl

Where tbenchx is the name of the test bench to be used.

For LDV rtl simulations use the command:

Make –f denali_LDV.mk TBENCH=tbenchx rtl

Where tbenchx is the name of the test bench to be used.

Simlar commands should be used for netlist simulations, for example :

Make –f denali.mk TBENCH=tbenchx net_max

2.8.6 Proposed Corrective action

The User guide has been updated for release r2p1

2.9 Write-Read-Write Sequence in MPMCDynamic Memory

2.9.1 Summary

If a Write-Read-Write occurs to the same address from the same port, the second write does not update the contents of the buffer at that address within that port.

2.9.2 Severity

Medium.

Whilst this an RTL issue, the conditions under which it happens are thought to happen very rarely.

2.9.3 Symptoms

The following sequence of events in simulation will show this error:

1. Write to any address from a given port.

2. Read the same address, making sure you do not cross a quad-word boundary in between, and also with an INCR transfer or any Burst Transfer which breaks in between. This would not happen if a single or Burst Read Transfers that does not break was carried out.

3. Write to the same address again.

4. Any read from this address produces the data written to that port in step 1 rather than that written in step3.

2.9.4 Affected items

MPMCBufCntl.vhd

2.9.5 Short term work around

Suggested solutions are:

1. Use r2p1 code. This issue does not exist in r2p1

2. Perform Reads from a different MPMC AHB Port.

3. Ensure that there is at least 100 clocks delay between the first read to the second write to same address from the same port.

4. Between any two writes to the same address, which happen from same port, if a read has to happen, then, make sure that the read will do a quad-word boundary cross. For example if any read in-between is carrying out a read from address 0x4, make sure it also reads 0x8, 0xC and 0x10.

5. An RTL fix for this issue can be provided by ARM in the REL1v0 Code. This would be a modified MPMCBufCntrl module which would have to be replaced into the existing design and the PL172 resynthesized. Contact support-primecell@arm.com for this code.

2.9.6 Proposed Corrective action

This issue does not exist in release r2p1

3 PL172 RELEASE R2P1 - ERRATA DETAILS

3.1 NAND Flash Control Software Drivers Not Included

3.1.1 Summary

The software drivers for NAND Flash are not included in Release 1v0.

3.1.2 Severity

Low

This is missing software that is not required for normal operation.

3.1.3 Symptoms

It will not be possible to use NAND Flash as standard at the current time.

3.1.4 Affected items

This is an issue with the software drivers.

3.1.5 Short term work around

It will not be possible to use NAND Flash as standard at the current time.

3.1.6 Proposed Corrective action

This product will not be able to support NAND Flash and thus there is no proposed corrective action.

3.2 Makefiles missing

3.2.1 Summary

Makefiles are missing from the following directories

PL172_VC/mpmc_pl172/verification

PL172_VC/mpmc_pl172/verification/vhdl

PL172_VC/mpmc_pl172/verification/verilog

PL172_VC/mpmc_pl172/verilog/netlist

PL172_VC/mpmc_pl172/vhdl/netlist

3.2.2 Severity

Low

3.2.3 Symptoms

Make commands for running verification tests will fail.

3.2.4 Affected items

Verification simulations.

3.2.5 Short term work around

3.2.6 Proposed corrective action

This has been corrected in release PL172r2p1-01rel0

3.3 Data Lost During SDRAM Write Access

3.3.1 Summary

A static access resulting in an error, followed by a dynamic write access results in the Arbiter block going to an illegal state, where a subsequent write is missed and this ultimately manifests as a loss of a write data.

3.3.2 Severity

High

This is an issue with the RTL.

3.3.3 Symptoms

AHB BurstA started on AHB_1 bus:

AHB BurstB started on AHB_1 bus

On memory interface, access for AHB Burst A takes place, however, the first word for AHB Burst B is missed

3.3.4 Affected items

MpmcAhbif.v/vhd

MpmcEndian.v/vhd

MpmcStTSM.v/vhd

MpmcMemBGnt.v/vhd

3.3.5 Proposed Corrective action

The RTL has been fixed in release PL172 r2p2-00ltd0.

3.4 Wrong Data Access in Case of SDRAM Read

3.4.1 Summary

During a static access to a write protected memory (which should result in an error and does) immediately followed by a dynamic read access, a spurious bypass condition occurs whereby the buffer is bypassed and data is fetched from the write-protected memory thus resulting in wrong data being accessed.

3.4.2 Severity

High

This is an issue with the RTL.

3.4.3 Symptoms

This issue was specifically found during a halfword access request. The MPMC provides the wrong data for a burst access starting at the address 0x23FE_403A and following addresses of this burst. The MPMC AHB interface still thinks it is in WORD access mode from the previous burst, but HALFWORD is currently requested.

3.4.4 Affected items

MpmcAhbif.v/vhd

MpmcEndian.v/vhd

MpmcStTSM.v/vhd

MpmcMemBGnt.v/vhd

3.4.5 Proposed Corrective action

The RTL has been fixed in release r2p2

3.5 No SDRAM Refresh During Long Static Memory Access

3.5.1 Summary

The MpmcMemBGnt block does not back off the static memory controller when the dynamic memory controller raises a request thus leading to the refresh cycles being missed.

3.5.2 Severity

High

This is an issue with the RTL.

3.5.3 Symptoms

After every read access, the static-state-machine in the MPMC is waiting in a turnaround cycle loop before relinquishing the external memory bus. However in this time, another read access from AHB is raised and this results in the static-machine never going to idle and hence there is a 'hogging' of the external bus which is preventing dynamic from giving out the refresh cycles.

3.5.4 Affected items

MpmcAhbif.v/vhd

MpmcEndian.v/vhd

MpmcStTSM.v/vhd

MpmcMemBGnt.v/vhd

3.5.5 Proposed Corrective action

The RTL has been fixed in release PL172 r2p2-00ltd0.

3.6 Arbitration locking to a low priority AHB port in some circumstances

3.6.1 Summary

The Mpmc does not rearbitrate to a higher priority AHB port when a large number of small burst transactions (burst <4) are submitted to a lower priority AHB port.

3.6.2 Severity

High

The Mpmc may not provide a low latency on the higher priority ports. Due to a lower priority AHB port locking arbitration.

3.6.3 Symptoms

A high priority port may not be provided with the expected latency, possibly causing the high priority AHB port master to over or underflow.

3.6.4 Affected Items

This is an issue with the RTL.

3.6.5 Short term work around

If possible minimise the number of small burst transactions. This could be reduced by writing software code to minimise the number of small write transactions which are not cache hits and to different pages of SDRAM. Such software algorithm modifications will both reduce the likelyhood of the arbitration issue, and improve system performance.

3.6.6 Proposed Corrective action

The RTL has been fixed in release PL172 r2p2-00ltd0.

3.7 A Write Following a Burst Transfer is Ignored When No Wait States Are Used [REJECTED]

This defect was incorrectly raised on the product and has been rejected.

It does not affect any revision of PL172.

3.8 Header file included that is not required

3.8.1 Summary

In the test code file pl172tst.c the file apint.h is called by an #include statement. This file is not required and will not be present in all environments so may cause a compiler error.

3.8.2 Severity

Low

3.8.3 Symptoms

Compiler error file apint.h not found.

3.8.4 Affected items

software

3.8.5 Short term work-around

Remove line from file pl172tst.c that calls file

#include "../apint/apint.h"

3.8.6 Proposed corrective action

The software has been fixed in release PL172 r2p2-00ltd0.

3.9 Data Read from memory and not from AHB write buffer if the buffers are full and the AHB write buffer does not contain a word of data.

3.9.1 Summary

This error occurs when all the buffers in the Mpmc are full and you perform a single or incrementing write which is not of total size word. For example 3 byte INCR or 1 half word. That write is stored in the AHB write buffer waiting for either another write to fill it and make it a word in size, or a buffer to become empty for it to write to. If you perform a read from the same address the Mpmc will read from memory and not from the AHB write buffer. If the write held in the AHB write buffer is a word in size this issue is not seen since the write buffer is redirected on the read. The data is not lost and will be written to memory later.

3.9.2 Severity

High

3.9.3 Symptoms

The data returned from a read will not be the data in the buffer but directly from memory. The data in the buffer will eventually be written to memory.

3.9.4 Affected items

RTL

3.9.5 Short term work-around

All writes must be of size word or make a word in length if you are reading it back immediately.

3.9.6 Proposed corrective action

The RTL has been fixed in release PL172 r2p2-00ltd0.

4 PL172 RELEASE R2P2-00LTD0 - ERRATA DETAILS

4.1 CKE state at power-up

ARM reference D/E 284509

4.1.1 Summary

At power-up the CKE signals are low. Some memory devices require both the DQM and CKE signals to be high otherwise they may drive the DQ signals and cause bus contention if multiple devices are present.

4.1.2 Severity

High - depending on the devices used

4.1.3 Symptoms

Bus contention could occur at power-up, until the initialisation process of the SDRAM has completed.

4.1.4 Affected items

RTL

4.1.5 Short term work-around

1) Use SDRAM devices that do not require the DQMs to be high at power-up or use only a single bank of devices.
2) The following RTL modification can be made. Please note that this change has not been validated at ARM, and so an extra test to test this change will have to be created. The existing test vectors will produce errors when simulating as they will be expecting the CKE to be LOW on reset.

- in MpmcDyIntRegBlk.v/vhd (line 2442): iDyCKEOut must be set (instead of cleared)when nPOR = '0' :

```
p_CKESeq : process (MPMCCLK, nPOR)
begin
    if (nPOR = '0') then
--    iDyCKEOut        <= "0000";   old line
     iDyCKEOut        <= "1111";  -- new line
  elsif (MPMCCLK'event and MPMCCLK = '1') then
    iDyCKEOut        <= NxtDyCKEOut;
  end if;
end process p_CKESeq;
```

4.1.6 Proposed corrective action

To be considered in a future release

4.2 An Obsolete Test Has Been Shipped and There are Issues with Some Tests and Log Files

ARM reference D/E 280959

4.2.1 Summary

1) There is a BankCross Test that exists in the verification/bustest directory that is not documented. This test is an obsoleted test that should not have been shipped. This test should be ignored.

2) The MpmcTrAhbif.vhd file does not compile using Modelsim version 5.6e. There is an improper concatenation within that file.

3) The MultiPortDiffComb8.c file does not compile as it has errors. More specifically, the variables LoBits and Bound are not declared properly.

4) The Mpmc.c file has a typo that causes the LatencyTest10.c to not be compiled.

5) The LatencyTest10 test does not pass.

6) The BusyInsertionTest does not pass.

7) Some of the RTL simulation log files are the incorrect log files

8) The verification/bustest README has some references to obsolete tests

4.2.2 Severity

Low – these are Testbench related issues affecting only two specific tests - MultiPortDiffComb8.c and

LatencyTest10.c

4.2.3 Symptoms

As summary for issues 1) to 4).

Issue 5) The simulation log file produces the following error:

** Error: MPMCTR49 : HREADY0 is low for more than the specified number of clocks

This error did not show up originally because the r2p2-00ltd0 version of this test was incorrectly running the "LatencyTest1.c"

Issue 6) The simulation log file produces the following error:

Error : Time 10957240 ns : MPMCTM19: Maximum time limit for the read access has exceeded

Instance bench.u3MpmcTrickMem.uMpmcTrMemRdWrCtl.p_RdWrComb

This error did not show up originally because the r2p2-00ltd0 version of this test was incorrectly running the "LatencyTest9.c" for modelsim and "LatencyTest1.c" for LDV.

Issue 8)The "SimMakefile_<SIMULATOR>.mk" should be

"DirSimMakefile_<SIMULATOR>.mk"

-> Tests - Arb2DyStWrRd and Arb3DyStWrRd - were not on the test-list with appropriate

tbench number

-> References are made to the following redundant tests:

-> WriteMisMatchTest

-> StBufFlushTest

-> DyBufFlushTest

-> MemModelFlashTest

-> MultiPortRandomTest2

Affected Items:

LatencyTest10.c Mpmc.c MpmcTrAhbif.vhd BusyInsertionTest.c

4.2.4 Affected items

3) MultiPortDiffComb8.c - Signals "Bound" and "LoBits" declared appropriately

(4) Mpmc.c - Typo "Latencytest10" changed to "LatencyTest10"

(5) MpmcTrAhbif.vhd - Number of zeros for HRDATA o/p for "MPMCTrStCS" register changed to [31:7]

4.2.5 Proposed corrective action

This issue has been fixed in the patch release r2p2-00ltd1

5 PL172 RELEASE R2P2-00LTD1 - ERRATA DETAILS

5.1 CKE state at power-up

ARM reference D/E 284509

5.1.1 Summary

At power-up the CKE signals are low. Some memory devices require both the DQM and CKE signals to be high otherwise they may drive the DQ signals and cause bus contention if multiple devices are present.

5.1.2 Severity

High - depending on the devices used

5.1.3 Symptoms

Bus contention could occur at power-up, until the initialisation process of the SDRAM has completed.

5.1.4 Affected items

RTL

5.1.5 Short term work-around

3) Use SDRAM devices that do not require the DQMs to be high at power-up or use only a single bank of devices.
4) The following RTL modification can be made. Please note that this change has not been validated at ARM, and so an extra test to test this change will have to be created. The existing test vectors will produce errors when simulating as they will be expecting the CKE to be LOW on reset.

- in MpmcDyIntRegBlk.v/vhd (line 2442): iDyCKEOut must be set (instead of cleared)when nPOR = '0' :

```
p_CKESeq : process (MPMCCLK, nPOR)
begin
   if (nPOR = '0') then
--    iDyCKEOut        <= "0000";   old line
     iDyCKEOut        <= "1111";  -- new line
  elsif (MPMCCLK'event and MPMCCLK = '1') then
    iDyCKEOut        <= NxtDyCKEOut;
  end if;
end process p_CKESeq;
```

5.1.6 Proposed corrective action

To be considered in a future release